ABJM $\mathfrak{su}(4)$ alternating chain: transfer matrices, ABJM point, discrete symmetries and selection rules

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All results below were obtained in wolfbook notebooks in `Paul/Mathematica/su4-alternating/`; the notebook that contains each result is given in brackets. Numerical checks are machine precision unless stated; "exact" means rational/symbolic arithmetic. Data files are in `data/`.

Contents

1 Chain, conventions and transfer matrices 1

2 The ABJM point 2

3 Highest-weight eigenstates and Q-functions 2

4 Discrete symmetries: LR, parity and their product 3

5 Structure constants of the note's configuration 3

6 Other three-point configurations 4

7 Factorised bras annihilated by the symmetry differences 4

8 Companion-twist SoV basis at four sites and BPS expansions 4

9 Open 6

1 Chain, conventions and transfer matrices

Chain. $2L$ sites ("L pairs"), alternating A F A F …, periodic. A sites carry the $\mathbf{4}$ ($\rho_A(E_{ij}) = e_{ij}$, group acts by g), F sites the $\bar{\mathbf{4}}$ ($\rho_F(E_{ij}) = -e_{ji}$, group acts by g^{-T}). In ABJM language: $\mathrm{tr}(Y^{A_1} Y^\dagger_{B_1} \cdots Y^{A_L} Y^\dagger_{B_L})$. Length $L = 2, 3, 4$ means $4, 6, 8$ sites and state spaces of dimension $256, 4096, 65536$. $P_{ab} = \sum e_{ij} \otimes e_{ji}$ (same-type sites), $K_{ab} = \sum e_{ij} \otimes e_{ij}$ (A–F pair; $4\times$ the singlet projector). $h = \frac{1}{2}$ throughout.

Fundamental transfer matrix. $\mathcal{L}_s(u) = \sum_{ij} e_{ij} \otimes \big((u-\theta_s)\delta_{ij} + h\,\rho_s(E_{ij})\big)$, i.e. $\mathcal{L}_A = u - \theta + hK$, $\mathcal{L}_F = u - \theta - hP$; $T = \mathcal{L}_1 \cdots \mathcal{L}_{2L}$, twist on the right, $t_1(u) = \mathrm{tr}\, T(u) G$. t_a are the quantum-minor fusions (rows antisymmetrised, column k at $u - (k-1)h$).

One-pass t_2 and t_3 [`dual-lax-t3.wb`, exact]. From the one-site quantum minors:

$$t_3(u) = \prod_s g_s^{(3)}(u)\ \mathrm{tr}\big[\bar{\mathcal{L}}_1(u)\cdots\bar{\mathcal{L}}_{2L}(u)\ \det G\, G^{-T}\big], \tag{1}$$

$$\bar{\mathcal{L}}_A = (u-\theta+h) - hP, \qquad g_A^{(3)} = (u-\theta-h)(u-\theta-2h),$$
$$\bar{\mathcal{L}}_F = (u-\theta-3h) + hK, \qquad g_F^{(3)} = (u-\theta)(u-\theta-h),$$

in the sign gauge $(-1)^k$ on the dual index; and

$$t_2(u) = \prod_s g_s^{(2)}(u)\ \mathrm{tr}\big[\mathcal{L}_1^{(2)}\cdots\mathcal{L}_{2L}^{(2)}\ \wedge^2 G\big], \quad \mathcal{L}_s^{(2)} = (u-\tilde{\theta}_s) + h\sum_{ab} D(e_{ab})\otimes\rho_s(E_{ab}), \tag{2}$$

D = derivation action on $\wedge^2\mathbb{C}^4$, $\tilde{\theta}_A = \theta$, $\tilde{\theta}_F = \theta + h$, $g_A^{(2)} = u-\theta-h$, $g_F^{(2)} = u-\theta$. Both checked identically against the quantum minors at $2L = 2, 4$ for companion, diagonal, generic and trivial twist. Relative A/F shift: 0 for t_1, h for t_2, $4h$ for t_3 (the earlier "naive" dual Lax had the right P/K structure but no relative shift). $\bar{t}_a$ denotes the reduced transfer matrices (without the scalar $\prod g_s$).

2 The ABJM point

[`abjm-chain.wb`, `abjm-bridges.wb`] Minahan–Zarembo Hamiltonian (arXiv:0806.3951; Bak–Rey arXiv:0807.2063), prefactor $\lambda^2/2$ dropped:

$$H_{\mathrm{MZ}} = \sum_{l=1}^{2L}\big(2 - 2P_{l,l+2} + P_{l,l+2}K_{l,l+1} + K_{l,l+1}P_{l,l+2}\big). \tag{3}$$

Hermitian, $\mathfrak{su}(4)$-invariant, invariant under the pair shift S. Spectrum at $L=2$: $E=0$ ($\times 84$, $[2,0,2]$ BPS), 4, 8 ($\times 120$), 12 ($\times 30$), 16 ($\times 20$), 20.

Result. For pair-periodic inhomogeneities (θ_A on A sites, θ_F on F sites), $[H_{\mathrm{MZ}}, \bar{t}_a(u)] = 0$ for $a = 1,2,3$ *iff* $\theta_A - \theta_F = 2h$ (exact at $L=2$ with u and $d = \theta_A - \theta_F$ symbolic: the gcd of all commutator coefficients is $\propto d - 2h$; numerically 10^{-15} at $L=3$). Moreover

$$U(u) = \bar{t}_1(\theta_F + u)\,\bar{t}_3(\theta_A - h + u), \qquad U(0) = \tfrac{1}{16}S, \qquad H_{\mathrm{MZ}} = U(0)^{-1}U'(0) + 12 \quad (L=2,\ \text{exact}). \tag{4}$$

In general $U(0) = c\,S$ with $c = h^4[x(-x-4h)]^2$ at $L=2$, $x = \theta_F - \theta_A$; it vanishes at $\theta_A - \theta_F \in \{0, 4h\}$.

Remark (superseded results). The earlier bridge computations used $\theta_A = \theta_F$, where $U(0) = 0$, the shift drops out of the t-algebra, H_{MZ} does not commute with the t's, and an odd-momentum HW partner is degenerate in t_1, t_2, t_3. All results below are at the ABJM point $\theta_A = 2h$, $\theta_F = 0$, untwisted and homogeneous unless stated.

3 Highest-weight eigenstates and Q-functions

States [`abjm-LR-symmetry.wb`]. All HW, cyclic ($S=1$) joint eigenstates of $\bar{t}_1, \bar{t}_3$: 6, 26, 152 at $L = 2, 3, 4$; all H_{MZ} eigenstates ($\leq 3.5\cdot 10^{-13}$), joint spectrum non-degenerate (gap ≥ 0.0065). Saved in `data/abjm-hw-states.wxf`; eigenvalue polynomials $\bar{t}_{1,2,3}(u)$ in `data/abjm-tau.wxf`.

Q-functions [`abjm-Qfunctions.wb`]. On the vacuum $T_{ii}|0\rangle = V_i|0\rangle$ with $V_1 = (u-\theta_A+h)^L(u-\theta_F)^L$, $V_2 = V_3 = (u-\theta_A)^L(u-\theta_F)^L$, $V_4 = (u-\theta_A)^L(u-\theta_F-h)^L$, and $\tau_a^{\rm vac}(u) = \sum_{i_1<\cdots<i_a} V_{i_1}(u)V_{i_2}(u-h)\cdots$, so $\sum_a(-1)^a\tau_a D^a = (1-V_1D)\cdots(1-V_4D)$ on the vacuum, $D = e^{-h\partial_u}$. Baxter equation in polynomial gauge:

$$\sum_{a=0}^{4}(-1)^a\,\tau_a(u)\,Q(u+h-ah)\prod_{b=a}^{3}V_4(u-bh)=0. \tag{5}$$

Q_1, Q_2, Q_3 = its polynomial solutions ordered by degree; $Q_{12} = \mathrm{Cas}(Q_1,Q_2)/\mathrm{Cas}_{\rm vac}$, $Q_{123} = \mathrm{Cas}(Q_1,Q_2,Q_3)/\mathrm{Cas}_{\rm vac}$. For all 184 states $\deg(Q_1,Q_{12},Q_{123}) = (K_1,K_2,K_3)$ with $K_2 = L - \frac{p+r}{2} - q$, $K_1 = \frac{L-r+K_2}{2}$, $K_3 = \frac{L-p+K_2}{2}$ for Dynkin $[p,q,r]$. Saved in `data/abjm-Qfunctions.wxf`.

4 Discrete symmetries: LR, parity and their product

[`abjm-LR-symmetry.wb`, `abjm-other-symmetries.wb`] Two operators commute with $H_{\rm MZ}$: $\mathcal{P}$ = (Weyl reversal $e_a \mapsto e_{5-a}$) $\circ$ (one-site shift), and $\mathcal{Q}$ = (signed Weyl g: $e_1\to e_4,\ e_2\to e_3,\ e_3\to e_2,\ e_4\to -e_1$) $\circ$ (reflection $s\mapsto 2L+1-s$). Numerically (10^{-16}):

$$\mathcal{P}\,\bar t_1(u)\,\mathcal{P}^{-1} = \bar t_3(u+h), \qquad \mathcal{Q}\,\bar t_1(u)\,\mathcal{Q}^{-1} = \bar t_1(2h-u), \qquad \mathcal{P}^2 = S, \quad \mathcal{Q}S\mathcal{Q}^{-1} = S^{-1}. \tag{6}$$

$\mathcal{P}$ is the ABJM version of 2603's LR map ($\mathbf{4}\leftrightarrow\bar{\mathbf{4}}$). Symmetric states, with the Q-function form verified exactly (fixed shifts) on all 184 states:

symmetry	eigenvalues	Q-functions
LR ($\mathcal{P}$)	$\tau_1(u) = \tau_3(u+h)$	$Q_1(u) \propto Q_{123}(u+h)$
P ($\mathcal{Q}$)	$\tau_1(u) = \tau_1(2h-u)$	$Q_1(u) \propto Q_{123}(2h-u)$
LP ($\mathcal{PQ}$)	$\tau_1(u) = \tau_3(3h-u)$	$Q_1(u) \propto Q_1(h-u)$

$\tau_2(u) = \tau_2(c-u)$ holds exactly for the P and LP states. Every non-LR state has an LR partner ($Q_1^{\rm partner} = Q_{123}(u+h)$). Counts:

L	states	LR	P	LP	all three	LR only	P only	LP only	none
2	6	4	6	4	4	0	2	0	0
3	26	8	18	12	8	0	10	4	4
4	152	24	60	40	20	4	40	20	68

$\mathcal{Q}$-eigenstates have eigenvalue $(-1)^p$ (in this sign convention).

5 Structure constants of the note's configuration

[`abjm-bridges.wb`, `abjm-LR-symmetry.wb`] Bra for bridge ℓ: first ℓ sites O_2 fields (A: e_1, F: e_4), the rest O_3 fields (A: e_1-e_4, F: e_4+e_1); $C_\ell = \langle B_\ell|\Psi\rangle/\|\Psi\|$.

$L=2$ at the ABJM point.

state	$E_{\rm MZ}$	nonzero C_ℓ
$[2,0,2]$ vacuum	0	all 1
$[0,2,0]$	16	—
$[1,0,1]$ parity pair	12, 12	$C_1 = C_3 = (1\mp i\sqrt{3})/\sqrt{48}$, $\lvert C_1\rvert = 1/\sqrt{12}$
singlet	20	$C_2 = -1/\sqrt{40}$
singlet	4	$C_2 = 1/\sqrt{24}$

The odd-bridge formula $\langle B_1|\Psi\rangle = \frac{1-z_4/(c_1\tau_1(\theta_1))}{1-z_4/z_1}\langle W_{-1}^L|\Psi\rangle$ holds through the untwisting limit at the ABJM point (10^{-11}–10^{-14}, generic inhomogeneous deformation, no shift operator needed).

When does C_ℓ vanish ($L = 2, 3, 4$). (i) R-charge: unless $\Psi \in [p, 0, p]$, and unless $L - p \le \ell \le L + p$. (ii) Bridge parity (kinematic, whole HW cyclic sector): if $L + p$ is odd, $C_\ell = 0$ unless $\ell \equiv L + p \pmod 2$. (iii) No LR selection rule: non-LR states have nonzero C_ℓ (e.g. the $L = 2$ adjoint pair). (iv) Isolated zeros at $L = 4$: the LR $[1, 0, 1]$ state at $E = 32$ (all ℓ) and the LR $[2, 0, 2]$ state at $E = 8$ (only even ℓ survive).

6 Other three-point configurations

[`abjm-3pt-configs.wb`] Two-block BPS bras: block 1 on ℓ sites with (a_1, f_1), block 2 on the rest with (a_2, f_2), $a_i \cdot f_i = 0$. Up to $GL(4)$ and rescalings only six classes; with block $1 = (e_1, e_4)$, block 2: generic $(e_1 - e_4, e_4 + e_1)$ (= the note's), $a_2 \cdot f_1 = 0$: $(e_2, e_1 + e_3)$, $a_1 \cdot f_2 = 0$: $(e_4 + e_2, e_3)$, both zero (e_2, e_3), $a_2 = a_1$: (e_1, e_3), $f_2 = f_1$: (e_2, e_4). Bridge starting on A or F. Whole multiplets probed by random global rotations.

LR. No configuration has an LR rule at $L = 3, 4$ (only an isolated $L = 2$ case). "Only non-LR states couple" occurs (8, 14, 24 configurations at $L = 2, 3, 4$) but not uniformly in L.

P: length-independent selection rules. With the *same polarisations at every* L, the following give $C = 0$ for every non-P state, verified at operator level on the cyclic sector ($\langle B|X_P(u)|r\rangle = 0$ for cyclic r, 10^{-16} at $L = 3, 4$; empty at $L = 2$ where every cyclic state is P-symmetric):

block 2	bridge ℓ	start
generic (note's W_{-1})	1 or $2L - 1$	A or F
$a_2 \cdot f_1 = 0$	1	A
$a_1 \cdot f_2 = 0$	1	F
both cross pairings zero, (e_2, e_3)	2 or $2L - 2$	A or F

Control: the note's configuration at $\ell = 2$ fails ($10^{-2.4}$). No uniform LP rule.

7 Factorised bras annihilated by the symmetry differences

[`abjm-LR-factorized.wb`, `abjm-other-symmetries.wb`] $X_{\rm LR}(u) = \bar{t}_1(u) - \bar{t}_3(u + h)$, $X_{\rm P}(u) = \bar{t}_1(u) - \bar{t}_1(2h - u)$, $X_{\rm LP}(u) = \bar{t}_1(u) - \bar{t}_3(3h - u)$.

- LR, $L = 2$ untwisted: $\ker X_{\rm LR}$ (left) has dim 136/256 ($X_{\rm LR}$ is linear in u); product states in it: only the BPS orbit. Twisted with $\Lambda = \mathrm{diag}(z_1, z_2, 1/z_2, 1/z_1)$ ($z_1 z_4 = z_2 z_3 = 1$, so the t_3 twist $\det \Lambda\, \Lambda^{-1}$ is the Weyl image of Λ): 36 LR-symmetric eigenstates, kernel dim 36, product states only $e_a \otimes e_{5-a}$ (vacuum and Weyl images); the note's W_a only for $a = 0$.
- P: *every* two-site state $(v \otimes w)^{\otimes L}$ is annihilated by $X_{\rm P}$ (10^{-16} at $L = 2, 3, 4$), the analogue of integrable two-site boundary states.
- LP: product solutions at $L = 2$ (arbitrary product states in a rotated SU(2)×SU(2) sector) fail at $L = 3, 4$.

8 Companion-twist SoV basis at four sites and BPS expansions

[`sov-companion-L4.wb`] Four sites A F A F, companion twist $G_c(z)$, generic θ. The 256 words

$$\langle a_1, k_2, a_3, k_4| = \langle 0|\, t_{a_1}(\theta_1 - h)\, t_1(\theta_2)^{k_2}\, t_{a_3}(\theta_3 - h)\, t_1(\theta_4)^{k_4}, \qquad a_i, k_i \in \{0, 1, 2, 3\}, \tag{7}$$

form a basis (exact rank 256), with factorised overlaps $\langle\Lambda|\Psi\rangle = \tau_{a_1}(\theta_1 - h)\tau_1(\theta_2)^{k_2}\tau_{a_3}(\theta_3 - h)\tau_1(\theta_4)^{k_4}\langle 0|\Psi\rangle$. Expansions $\langle B| = \sum_\Lambda c_\Lambda \langle\Lambda|$ of BPS states $v \otimes w \otimes v \otimes w$, $v \cdot w = 0$ (number of nonzero c_Λ):

BPS state	$\#\{c_\Lambda \neq 0\}$
vacuum	1
F sites only, $v = e_1$, $w \in e_1^\perp$ general	9 $(a_1 = a_3 = 0,\ k_2, k_4 \leq 2)$
F only, $\exp(\kappa E_{34})$	4 $(k \leq 1$; not a product, F–F exchange term$)$
F only, $\exp(\kappa E_{24})$	5
A only, $\exp(\kappa E_{12})$	13 $(a \leq 2,\ k \leq 1)$
A only, $w = e_4$, $v \in e_4^\perp$ general	56
SU(2)×SU(2) sector	37
torus $\mathbb{C}[G_c^T]$	134–236
generic	256

Single-site (non-BPS) rotations site $1 + \kappa e_4$ and site $2 + \kappa e_3$ have two-term expansions.

At the ABJM point pair by pair. With $\theta_{A,n} = \theta_{F,n} + 2h$ and generic $\theta_{F,n}$, the words above degenerate: at $u = \theta_A - h = \theta_F + h$ the fused t_3 vanishes through its scalar factor, and even with the reduced $\bar{t}_2, \bar{t}_3$ the same-pair F Lax is $h(1 - P)$, giving rank $169 = 13^2$ (13 per pair; same for $u_A = \theta_A, \theta_A \pm h, \theta_A - 2h$). A word basis (rank 256) is restored with the A-site words at $u_A = \theta_A - 3h$ or $\theta_A + 2h$, using $\bar{t}_2, \bar{t}_3$, with unchanged BPS sparsity — *but it is not an SoV basis.*

Localisation on sites 1, 2 (generic θ, exact). Among the 16 site-1,2 words $\langle 0|t_a(\theta_1 - h)t_1(\theta_2)^k$, exactly $\{1, t_3(\theta_1 - h)\} \times \{1, t_1(\theta_2)\}$ are localised (sites 3,4 at the vacuum), and the constants factorise:

$$\langle 0|\big(1 + \alpha\, t_3(\theta_1 - h)\big)\big(1 + \beta\, t_1(\theta_2)\big) = (e_1 + \alpha k_3 e_4) \otimes (e_4 + \beta k_1 e_3) \otimes e_1 \otimes e_4, \tag{8}$$

a two-parameter product rotation of sites 1, 2 only. The localised span is 5-dimensional; the extra element $\langle 0|[t_1(\theta_2)^2 + c\, t_1(\theta_1 - h)t_1(\theta_2)]$ is entangled on sites 1, 2 ($e_1 \otimes e_2 + c' e_2 \otimes e_3$, reaching $e_4^T G_c^2$ on site 2) and adds no new product states.

Sites 1, 2, 3. Among the 64 words with $k_4 = 0$, the localised single words are $\{1, t_3(\theta_1 - h)\} \times \{1, t_1(\theta_2)\} \times \{1, t_3(\theta_3 - h)\}$ (localised span 15-dim), but $t_3(\theta_3 - h)$ on $\langle 0|$ gives an F_2–A_3 singlet plus an exchange term. Localised product states form exactly two families: (A) $(e_1 + a e_4) \otimes (e_4 + c e_3) \otimes e_1 \otimes e_4$ and (B) $(e_1 + a e_4) \otimes e_3 \otimes (e_1 + b e_4) \otimes e_4$ (arbitrary a, b; site 3 rotates only when site 2 is fully flipped); (B) $= \langle 0|\, t_1(\theta_2) \times$(degree ≤ 1 in $t_3(\theta_1 - h)$, $t_3(\theta_3 - h)$, with an exchange correction). No three-parameter product rotation of sites 1, 2, 3.

SoV vacuum. Ryan–Volin's $\langle 0|$ (arXiv:2002.12341, (4.12)) is the lowest-weight GT vacuum, $\langle 0|T_{j1}(u) = \delta_{j1}\nu(u)\langle 0|$ (3.19) applied recursively through the embeddings, i.e. killed by all strictly lower-triangular $T_{ij}(u)$, $i > j$, and eigen of the T_{ii} (untwisted monodromy; our $\mathcal{L}$ and RV's agree with $\pi = -\rho^T$ and site order reversed). The solution is unique and equals our $\langle 0| = e_1 \otimes e_4 \otimes e_1 \otimes e_4$ (exact, generic and ABJM θ). RV's genericity assumption (2.9), $\theta_\alpha - \theta_\beta \notin \hbar\mathbb{Z}$, fails at the ABJM point.

B-operator check. Ryan–Volin $B(u) = \sum_{k_1} \sum_{K_2} T_{k_1,1}(u)\, \mathrm{qmin}_{K_2;(1,k_1+1)}(u+h)\, \mathrm{qmin}_{(1,2,3);(1,K_2+1)}(u+2h)$ from the untwisted T, $[B(u), B(v)] = 0$. At generic θ all 256 words ($u_A = \theta_A - h$) are B-eigenbras (exact): a genuine SoV basis, so the sparse expansions above are SoV expansions. At $\theta_A = \theta_F + 2h$ only the 16 F-only words are B-eigen, and B itself is degenerate: the number of distinct eigenvalues is 256 for $d = \theta_A - \theta_F \in \{0, 4h, \text{generic}\}$ but $169 = 13^2$ for $d \in \{h, 2h, 3h\}$ (collision of the separated variables of the two sites of a pair). The ABJM point therefore needs either a limit $d \to 2h$ of the generic SoV basis or a different B.

9 Open

- Mechanism behind the $\ell = 1$ (and both-zero $\ell = 2$) P rules; test $L = 5$.
- Two-block non-BPS families ($v \cdot w \neq 0$) for general ℓ; transfer-matrix-built bridges as in 2603.
- Twisted versions (2603 works at generic twist with $\omega = 1$); LR-compatible non-diagonal twists.
- SoV: companion-twist SoV basis at $2L = 4$ and SoV expansions of BPS products (next).

References. B. Bargheer, C. Bercini, B. Lefundes, P. Ryan, arXiv:2603.26617; J. A. Minahan, K. Zarembo, arXiv:0806.3951; D. Bak, S.-J. Rey, arXiv:0807.2063; P. Ryan, D. Volin, arXiv:1810.10996, arXiv:2002.12341.